How Parched the Sites Soil Series Really Is

Updated by Cody Selig on October 18, 2026

How much available plant water is in the root zone of the Sites soil series?

The Calculations of the available water holding capacity in this soil series are backed up from using the knowledge of textures found in its horizons. Although the horizons are not very differing in texture, showing signs of only loams and clays, it does show many more attributes regarding structure, rock percentage, and pH levels. This is important because these traits can affect the way we calculate the PAW within any given soil series. In this example, the estimates of field capacity, wilting point, and bulk density only relied on the texture of the soil using the CDM Smith Soil Hydraulic Properties calculator. With the data we can calculate that the Sites soil series had an estimated amount 18.31% of available water holding capacity. However, this number can change with the presence of rock percentage. Rock percentage being present in this calculation would increase the bulk density of any given horizon, the higher the amount of rock fragments, the higher the density of the soil. Furthermore, organic matter can also dictate the PAW for each horizon. Most organic matter can absorb a lot of PAW, making it unable to trickle down into the lower horizons, for example, “acts similar to a sponge, absorbing and holding up to 90% of its weight in water” (Agri Tec Agronomy Team 2022). 90% is a massive amount, especially when you think of the impact that has on bulk density, field capacity, wilting point. Although, organic matter of course varies depending on location and many other factors, and even sometimes helps retain a lot of water so it doesn’t drain to lower horizons where the tension would be too high to ever be used by plants. Overall, the PAW in this series can be estimated by knowing the textures of the soil, but by knowing the rock percentage and impact of the organic matter we can further calculate how much water is available in this soil.

Hrz	thickness(	texture	clay%	sand%	Fc	wilt point	bd(g/cm)	final PAW
A1	17	L	20	40	0.379	0.224	1.407	0.137
A2	18	L	20	40	0.379	0.224	1.407	0.137
BAt	18	L	20	40	0.379	0.224	1.407	0.137
Bt1	20	C	60	15	0.49	0.353	1.198	0.137
Bt2	61	C	60	15	0.49	0.353	1.198	0.137
BCt	41	C	60	15	0.49	0.353	1.198	0.137

Soil Series: Sites Location: California 10/18/26

tPAW: = 18.31% AWHC

How much plant available water is in the root zone of this favorite soil?

		thickness (cm)			
8-25	A_1	17cm	L	sPAW = .155	2.15%
43	A_2	18 cm	L	.155	2.28%
61	BAt	18 cm	L	.155	PAW 2.28%
81	B_{t1}	20 cm	C	.137	1.92%
142	B_{t2}	61 cm	C	.137	5.75%
183	BCt	41 cm	C	.137	3.93%
198cm	Cr (no roots)	15cm		total PAW% =	18.31% AWHC

rootzone = 175cm thickness of where roots reach

Depth · Density $\left(\frac{FC-WP}{rootzone}\right)$ = available water holding capacity

A_1 $17 \cdot 1.407\left(\frac{.379}{175} - \frac{.224}{175}\right) = 2.15\%$

.0022 - .0013 = (.0009) $17 \cdot 1.407 = .0215271 \approx .0215$

A_2 $18 \cdot 1.407\left(\frac{.379-.224}{175}\right) = (.0009)\,18 \cdot 1.407 = .0228 = 2.28\%$

BAt = same as A_2

B_{t1} $20 \cdot 1.198\left(\frac{.49-.353}{175}\right) = (.0008)\,20 \cdot 1.198 = .0192 = 1.92\%$

B_{t2} $60 \cdot 1.198\left(\frac{.49-.353}{175}\right) = (.0008)\,60 \cdot 1.198 = .0575 = 5.75\%$

BCt $41 \cdot 1.198\left(\frac{.49-.353}{175}\right) = (.0008)\,41 \cdot 1.198 = .0393 = 3.93\%$